CCPN

CcpNmr Analysis Version 3

How To:

Import NMRtist results into CcpNmr Analysis version 3

NMRtist

www.nmrtist.org

```
data_nef_file_from_cyana

# NMR meta data

save_nef_nmr_meta_data

    _nef_nmr_meta_data.sf_category          nef_nmr_meta_data
    _nef_nmr_meta_data.sf_framecode         nef_nmr_meta_data
    _nef_nmr_meta_data.format_name          nmr_exchange_format
    _nef_nmr_meta_data.format_version       1.1
    _nef_nmr_meta_data.program_name         CYANA
    _nef_nmr_meta_data.program_version      4.0
    _nef_nmr_meta_data.creation_date        2024-11-26T23:48:55
    _nef_nmr_meta_data.uuid                 CYANA-2024-11-26T23:48:55-5385537942

    loop_
        _nef_program_script.program_name
        _nef_program_script.script_name
        _nef_program_script.script

REMARK Contents:4
;
nef_molecular_system                            151 residues
nef_nmr_spectrum_C13HSQC_hmqc_@FLYA_asn         707 peaks with 450 assignments
nef_nmr_spectrum_C13NOESY_13cnoesyarom_@FLYA_asn  3948 peaks with 213 assignments
nef_nmr_spectrum_CBCANH_cbcanh_@FLYA_asn        801 peaks with 208 assignments
nef_nmr_spectrum_CBCAcoNH_cbcaconh_@FLYA_asn    427 peaks with 108 assignments
nef_nmr_spectrum_CcoNH_ccconh_@FLYA_asn         622 peaks with 151 assignments
nef_nmr_spectrum_HBHAcoNH_hbhacohn_@FLYA_asn    598 peaks with 60 assignments
nef_nmr_spectrum_HCCHTOCSY_hcchtocsy_@FLYA_asn  1530 peaks with 359 assignments
nef_nmr_spectrum_HNcaCO_hncaco_@FLYA_asn        433 peaks with 0 assignments
nef_nmr_spectrum_N15HSQC_hsqc_@FLYA_asn         255 peaks with 129 assignments
nef_nmr_spectrum_N15NOESY_noesyhsqc_@FLYA_asn  6416 peaks with 918 assignments
;
    stop_

save_


# Molecular system

save_nef_molecular_system

    _nef_molecular_system.sf_category       nef_molecular_system
    _nef_molecular_system.sf_framecode      nef_molecular_system

    loop_
        _nef_sequence.index
        _nef_sequence.chain_code
        _nef_sequence.sequence_code
        _nef_sequence.residue_name
        _nef_sequence.linking
        _nef_sequence.residue_variant

        1 .   1 SER  start  .
        2 .   2 MET  middle .
        3 .   3 VAL  middle .
        4 .   4 SER  middle .
        5 .   5 PRO  middle .
        6 .   6 GLU  middle .
        7 .   7 GLU  middle .
        8 .   8 ALA  middle .
        9 .   9 VAL  middle .
```

Chemical Shift Table

ChemicalShiftList CL:shifts

Value (ppm)	NmrAtom
5.648	NA:A.1.HIS.HA
4.028	NA:A.1.HIS.HBx
4.369	NA:A.1.HIS.HBy
8.662	NA:A.1.HIS.HD1
7.400	NA:A.1.HIS.HD2
7.955	NA:A.1.HIS.HE1
56.113	NA:A.2.MET.CA
31.486	NA:A.2.MET.CB
27.054	NA:A.2.MET.CE
31.501	NA:A.2.MET.CG
8.996	NA:A.2.MET.H
4.118	NA:A.2.MET.HA
3.102	NA:A.2.MET.HBx
3.240	NA:A.2.MET.HBy
3.102	NA:A.2.MET.HGx
3.239	NA:A.2.MET.HGy
113.996	NA:A.2.MET.N
55.577	NA:A.3.ARG.CA
31.252	NA:A.3.ARG.CB
43.227	NA:A.3.ARG.CD
27.544	NA:A.3.ARG.CG
8.448	NA:A.3.ARG.H
4.376	NA:A.3.ARG.HA
1.615	NA:A.3.ARG.HBx
1.844	NA:A.3.ARG.HBy
3.134	NA:A.3.ARG.HD%
1.615	NA:A.3.ARG.HG%
121.984	NA:A.3.ARG.N
53.722	NA:A.4.GLN.CA
29.259	NA:A.4.GLN.CB

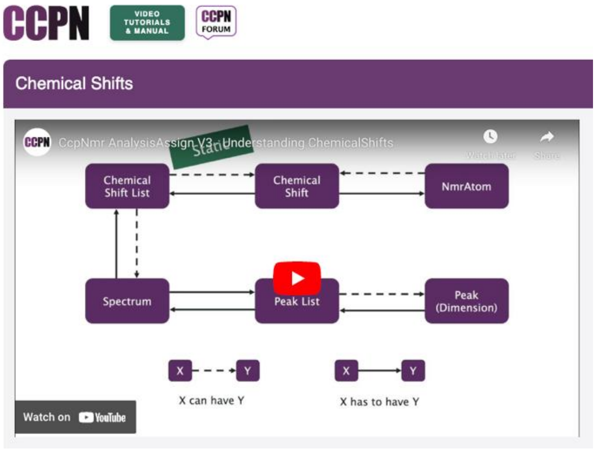
Introduction

This tutorial is designed to give an overview of the steps involved in importing the results from NMRtist peak picking and chemical shift assignments into a CcpNmr Analysis V3 project.

This tutorial does not give any instructions about the NMRtist server itself; users should use the tutorials and guidelines available on the NMRtist website: https://nmrtist.org/

Additionally, we recommend users familiarize themselves with a few concepts from the CcpNmr Analysis manual:

1. Handling chemical shifts in relation to spectra:
https://www.ccpn.ac.uk/manual/v3/ChemicalShifts.html



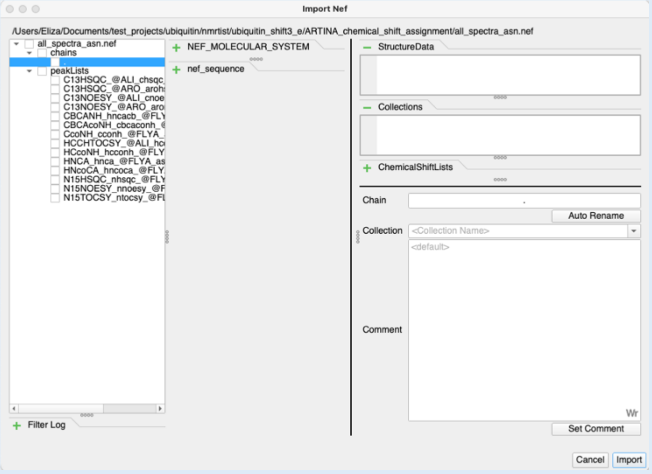
2. Basic information about the NEF format
https://www.ccpn.ac.uk/manual/v3/NEF.html

Tutorial Summary

Main Steps

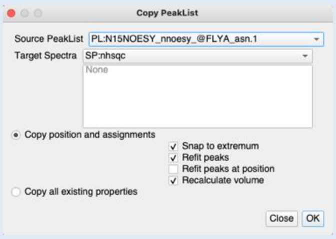
- Open project
- Import NMRtist Peak Lists from NEF file
    - Update Chain Code
    - Name Chemical Shift List for imported Peak Lists

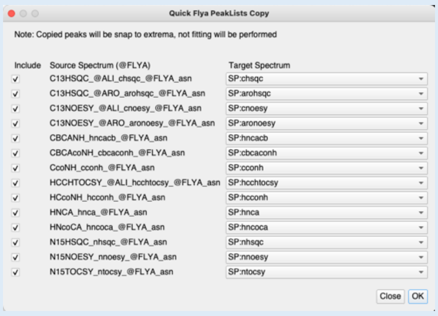
1



- Copy Peak Lists to existing Spectra

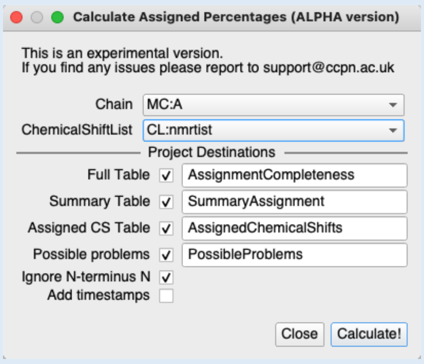
2





- Inspect Peak Lists and Chemical Shifts

3

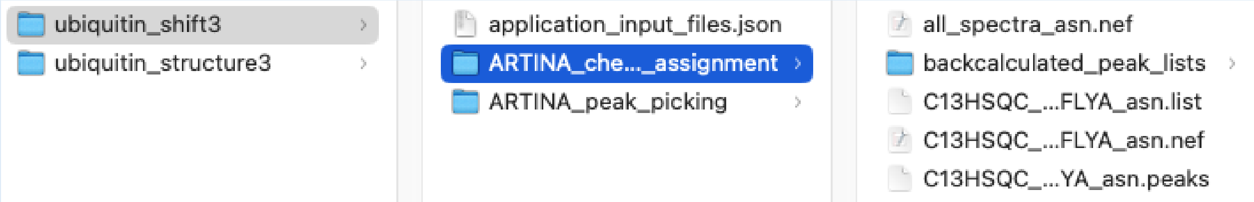


1 NEF files provided by NMRtist

Different NMRtist Applications will produce NEF files with slightly different content.

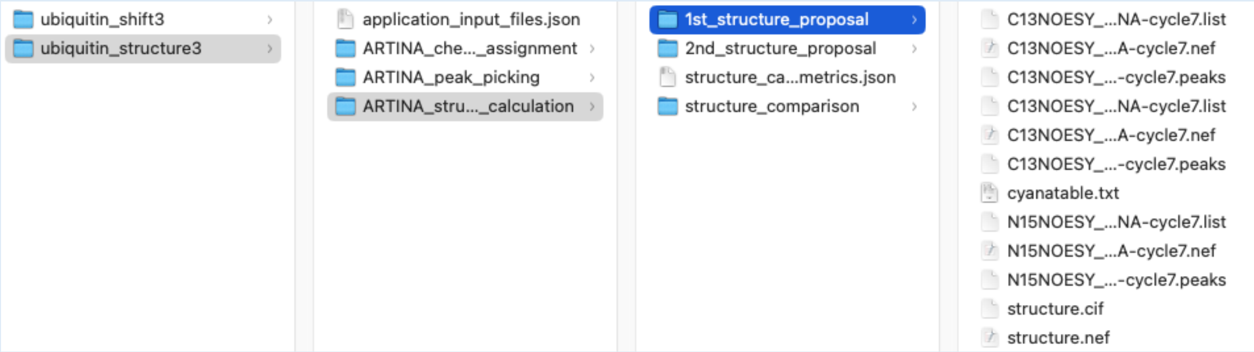
In this document we will focus on importing peak lists, which contain assigned peaks. Those will give a generated chemical shift list.

The **ARTINA: chemical shift assignment** application gives two folders (May 2025):



ARTINA_chemical_shift_assignment folder contains:
- **all_spectra_asn.nef** file with all peak lists (spectra), no chemical shift list
- **shifts.nef** and **shifts-strong.nef** with no peak lists
- NEF files with individual peak lists (spectra)
- backcalculated_peak_lists folder with NEF files with individual peak lists (spectra)

The **ARTINA: structure determination** application gives three folders:



ARTINA_structure_calculation/1st_structure_proposal (or 2nd_structure_proposal) contains:
- **structure.nef** with chemical shift list and distance restraints (those can differ between 1st and 2nd proposal), but no peak lists.

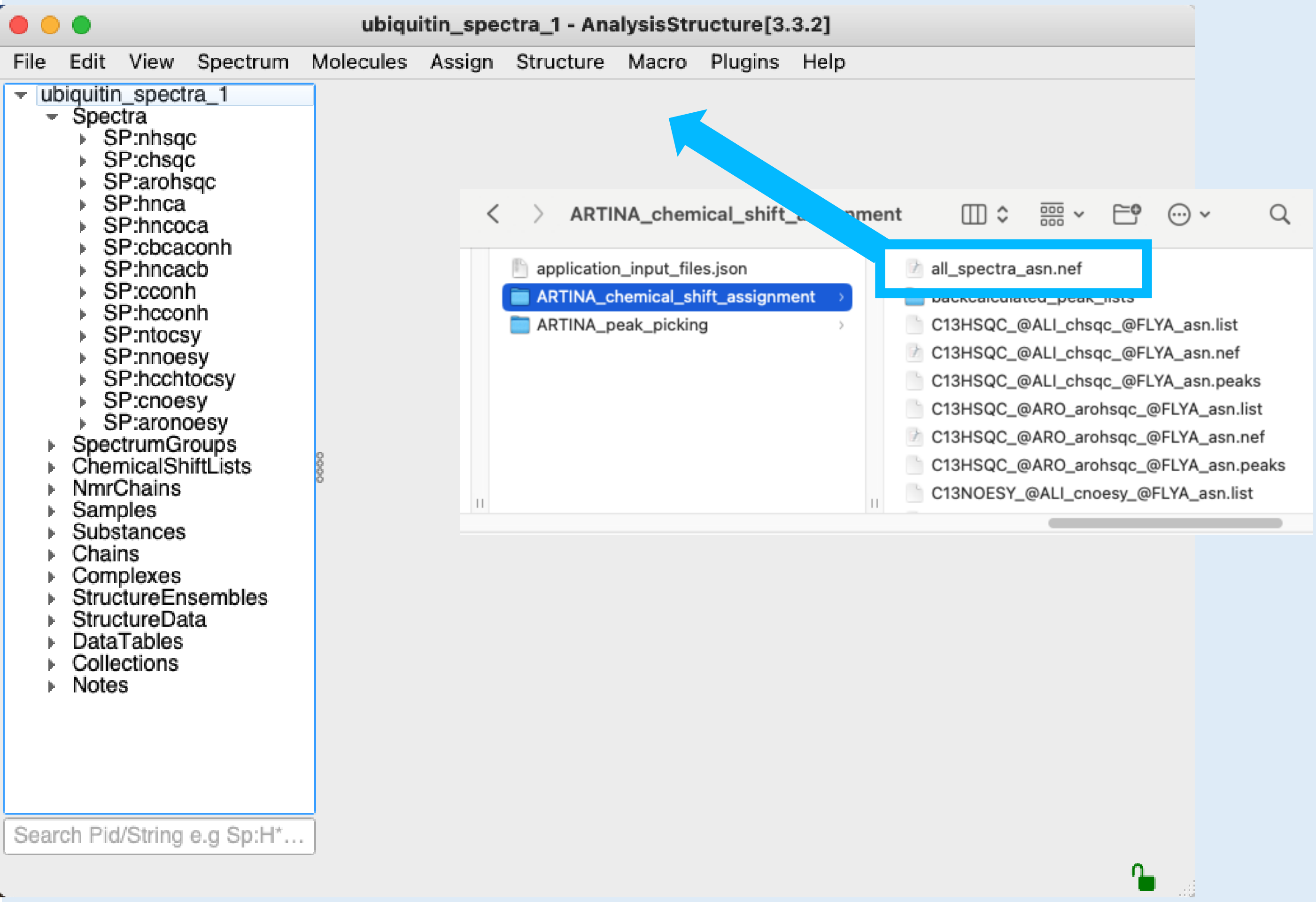
Peak lists can be found as individual NEF files in the folder **ARTINA_chemical_shift_assignment**.

Note that there are small differences between the chemical shifts loaded from files such as **shifts.nef**, **structure.nef** and **all_spectra_asn.nef**.

All Artina's NEF files contain sequence information.

1 Importing results into v3 project

ubiquitin_spectra_1.ccpn



All files needed for this tutorial are located in the **nmrtist_import** directory.

The CCPN project used contains all the spectra (with Experiment Types already set up) which were used for an NMRtist run. For information how to do load spectra and set experiment types, please see our CcpNmr Analysis Basic Tutorial.

1A Open CCPN Analysis V3.

CcpNmr projects are saved as directories with the **.ccpn** extension.

- Find the project **ubiquitin_spectra_1.ccpn** in the tutorial data directory and drag & drop it into the program.
- Find and unpack results from the NMRtist.org website: **nmrtist/ubiquitin_shift3.zip**

  You should now have a folder with two new directories and a .json file.
- Go to the **ARTINA_chemical_shift_assignment** folder.
- Go to **File / Import > Nef File** and select the **all_spectra_asn.nef** file.

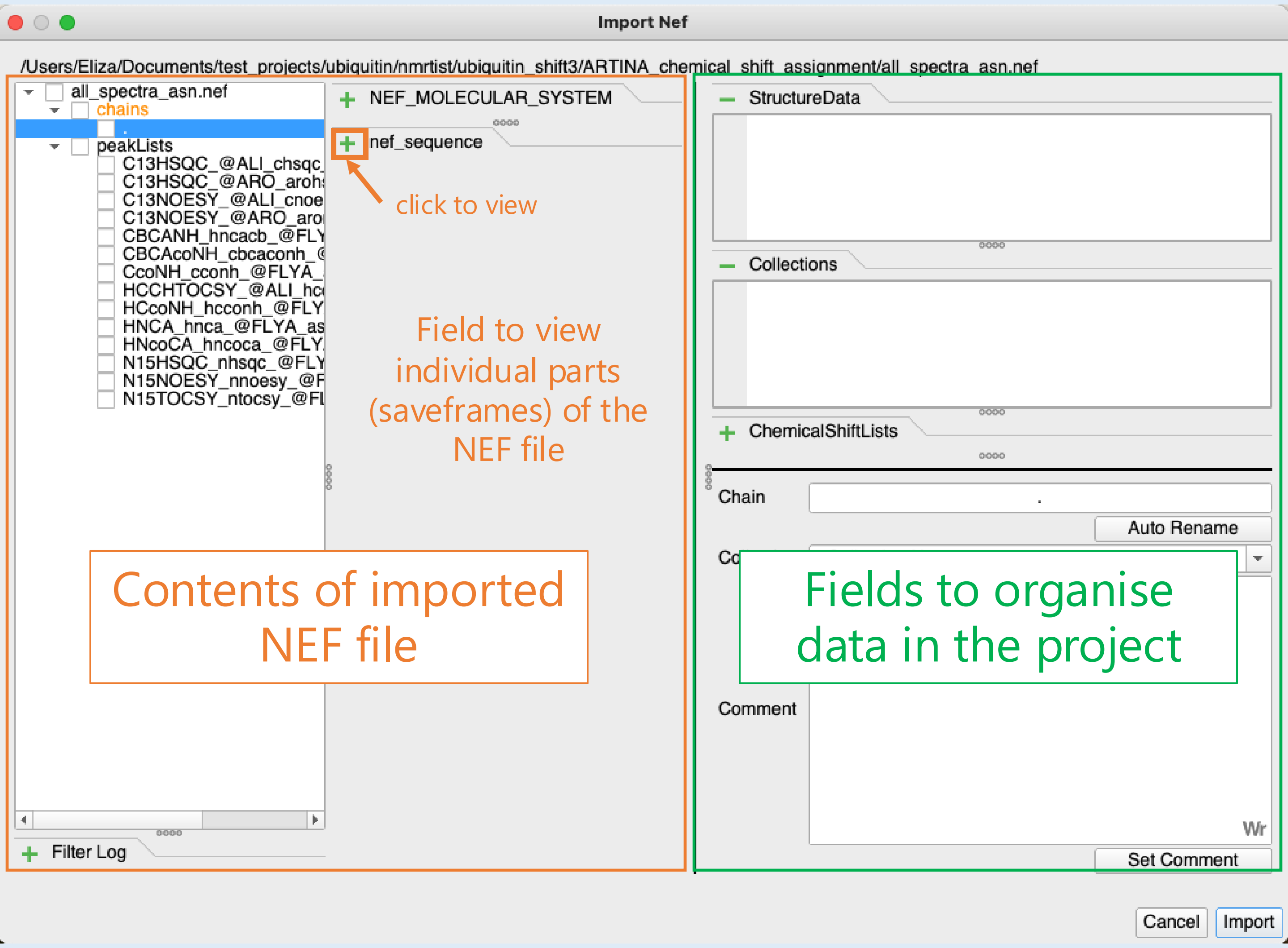
OR

- Drag & Drop the **all_spectra_asn.nef** file into the Drop Area of the program.
- On the prompt click **Import**.
- You will see the Import NEF pop up window opening.

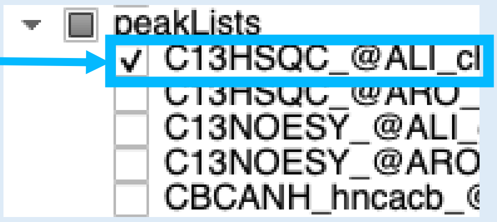
1

Importing results into v3 project

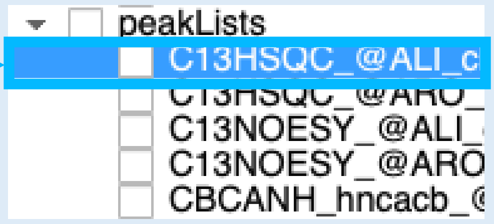
ubiquitin_spectra_1.ccpn



Ticked for import



Selected for edit



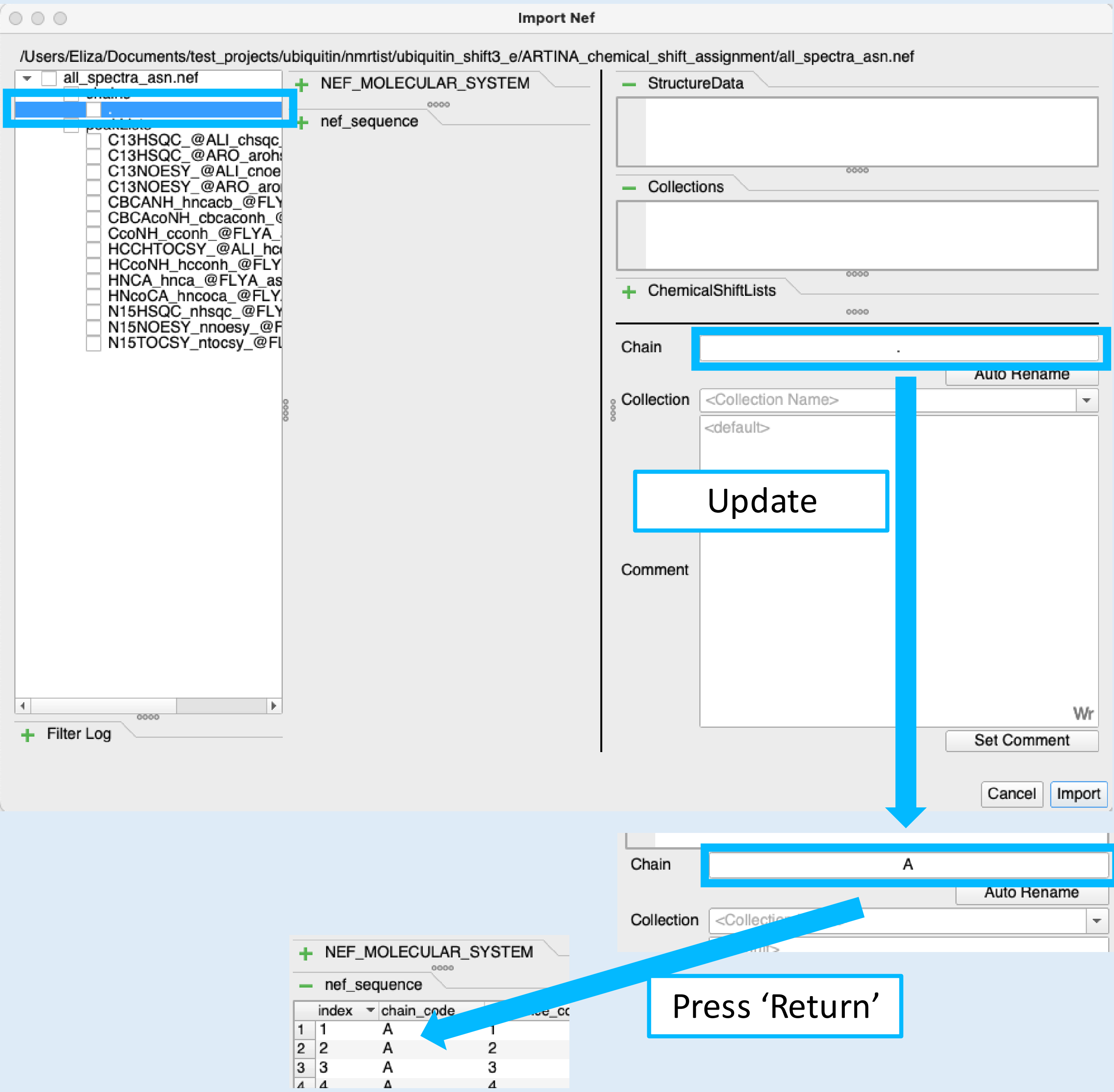
1B **Loading and updating chain from NEF**

- Items chosen for import must be ticked
- You can select item(s) for editing by clicking on the item (the background will be highlighted in blue). If you start to edit an item, it will be ticked by default.
- You can view/inspect the contents of items by clicking on the green plus sign in the middle of the pop-up window.

Importing results into v3 project

ubiquitin_spectra_1.ccpn

NMRtist provides a NEF file in which no symbol is used for the Chain Code.
We recommend changing it to a capital letter, as is common practice elsewhere.



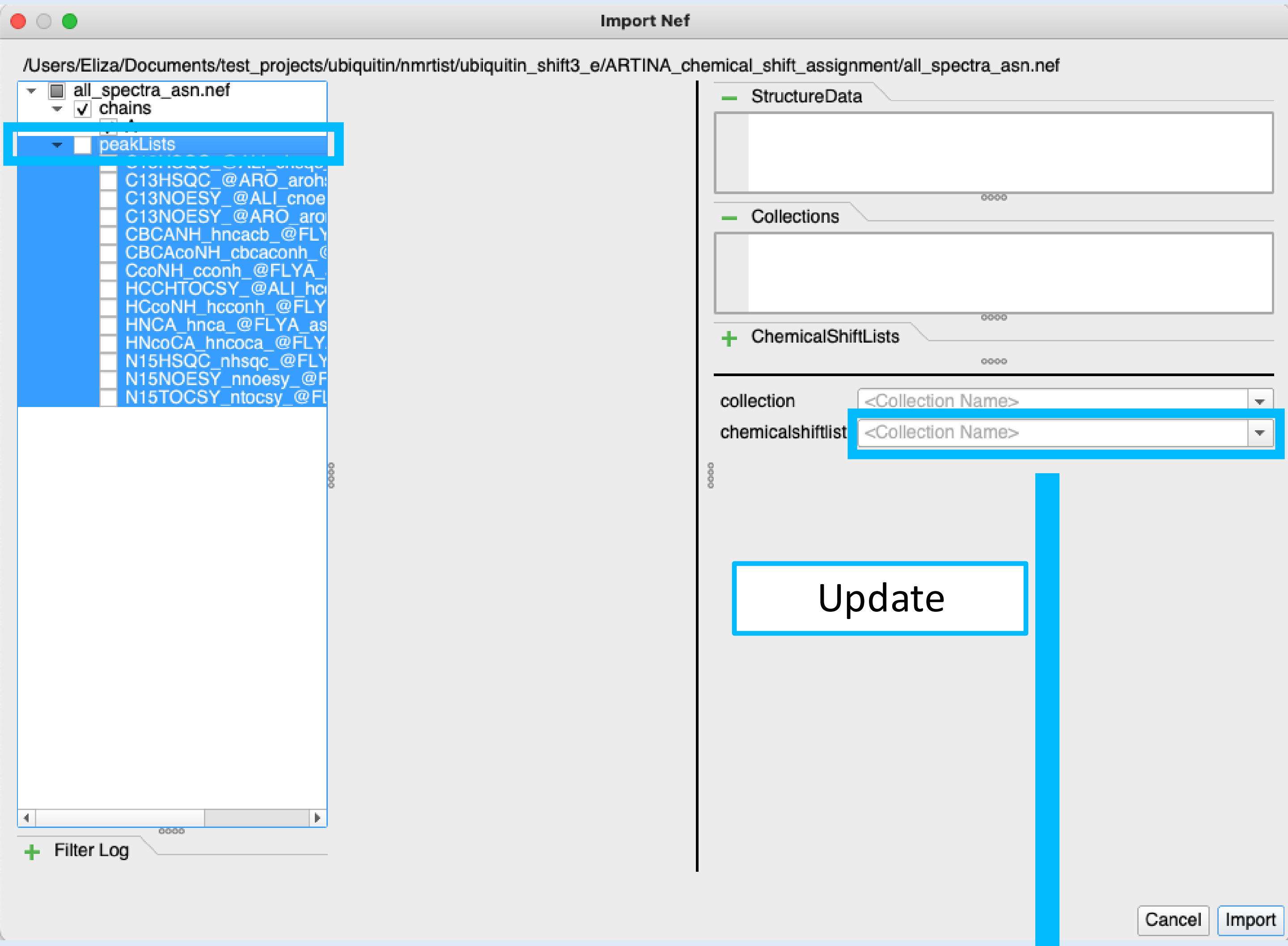
1c Loading and updating chain from NEF *(continued)*

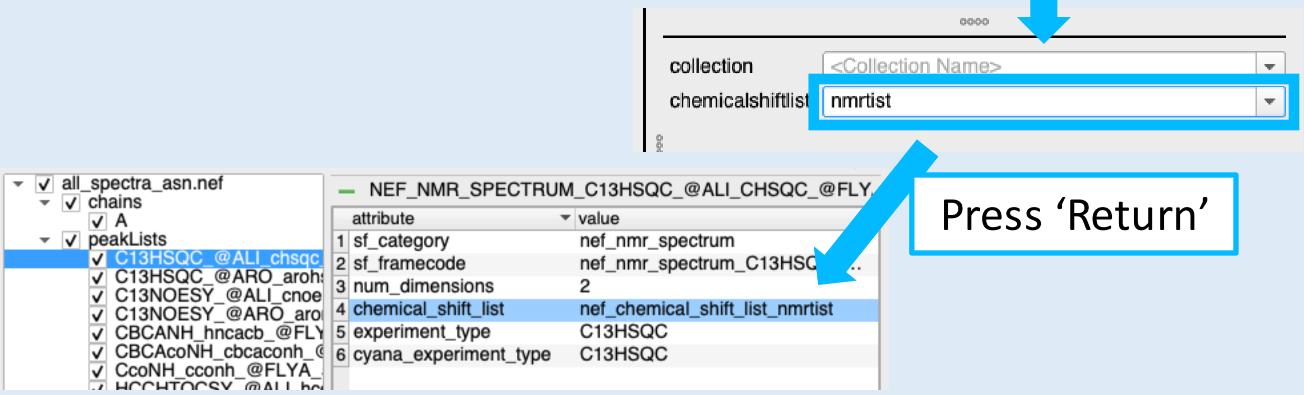
- Tick **‘.’**, this will highlight the chain section ready for editing.
- On the right side in the **Chain** field change the name to **‘A’** (or another name of your choice). **Press ’Return’ to confirm.**
- If you expand the section on the left side of the window you can see the Chain Code is now updated.

1 Importing results into v3 project

ubiquitin_spectra_1.ccpn

NEF files imported from the **ARTINA_chemical_shift_assignment** folder do not have a chemical shift list associated with the peak lists. This should be updated during import.





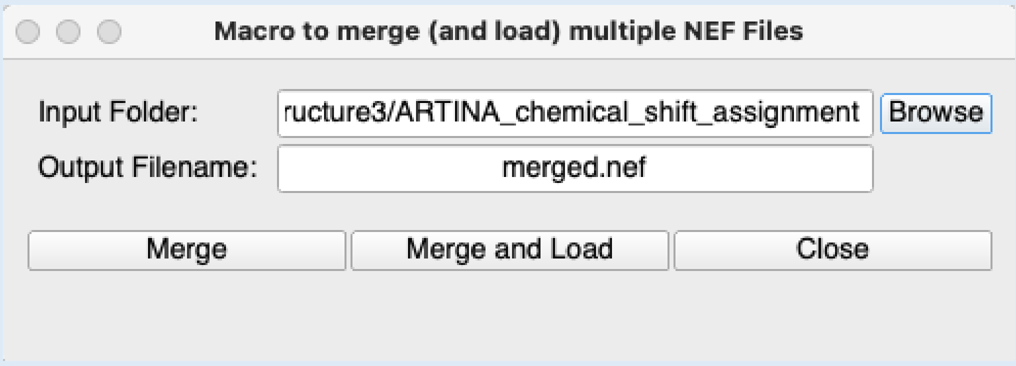
1D Naming chemical shift list for imported peak list

- Click on **peakLists** on the left-hand side.
  This will highlight all spectra ready for editing.
- On the right-hand side in the **ChemicalShiftList** field change the name to **nmrtist** (or another name of your choice).
- Press **Return** to confirm.
- If you expand the section on the left-hand side of the window for any of the spectra, you can see the chemical shift list name is now updated.
  All PeakLists should be automatically ticked for import
- Click **Import, which** may take few minutes, depending on the number of spectra.

1 Importing results into v3 project

ubiquitin_spectra_1.ccpn

As mentioned on page 3, the **ARTINA: structure determination** application will not generate a NEF file that includes peak lists. If you want to import a NEF file from this application, you can run a CCPN Macro that will create and load such a NEF file by merging this NEF file with others .



1E Merging and loading a NEF file from ARTINA: structure determination folder

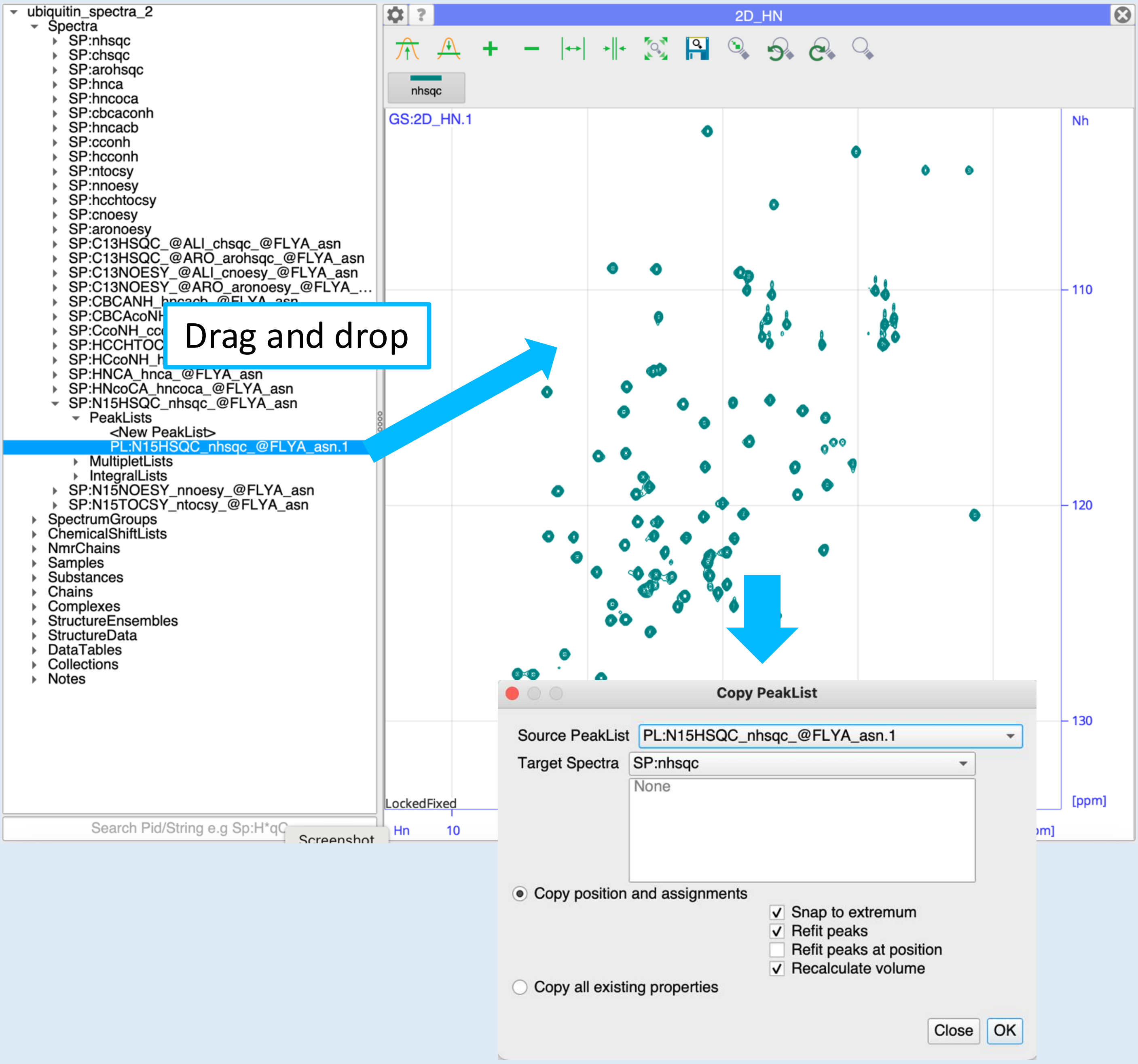
- Go to **Macro > Run CCPN Macros > mergeMulitiNef**
- In the resulting popup type in the path to the folder containing the files you wish to merge or navigate to the folder using the **Browse** button.
  From the tutorial data this would be **nmrtist_import / nmrtist / ubiquitin_structure3 / ARTINA_structure_calculation / 1st_structure_proposal** folder.
- Update the name of the resulting file if you wish.
- Click **Merge and Load,** in the resulting popup click **Import**
- *Alternatively:* Click **Merge** and load NEF as described in point **1A**
- Now continue with steps **1B – 1D**

Note: This macro can also be used to merge NEF files from the **ARTINA_chemical_shift_assignment / backcalculated_peak_lists** folder if desired

2 Coping peak lists between spectra

ubiquitin_spectra_2.ccpn

Peak lists will be imported as separate spectra associated with separate chemical shift list. The next steps is to copy all or selected peak lists to their target spectra.

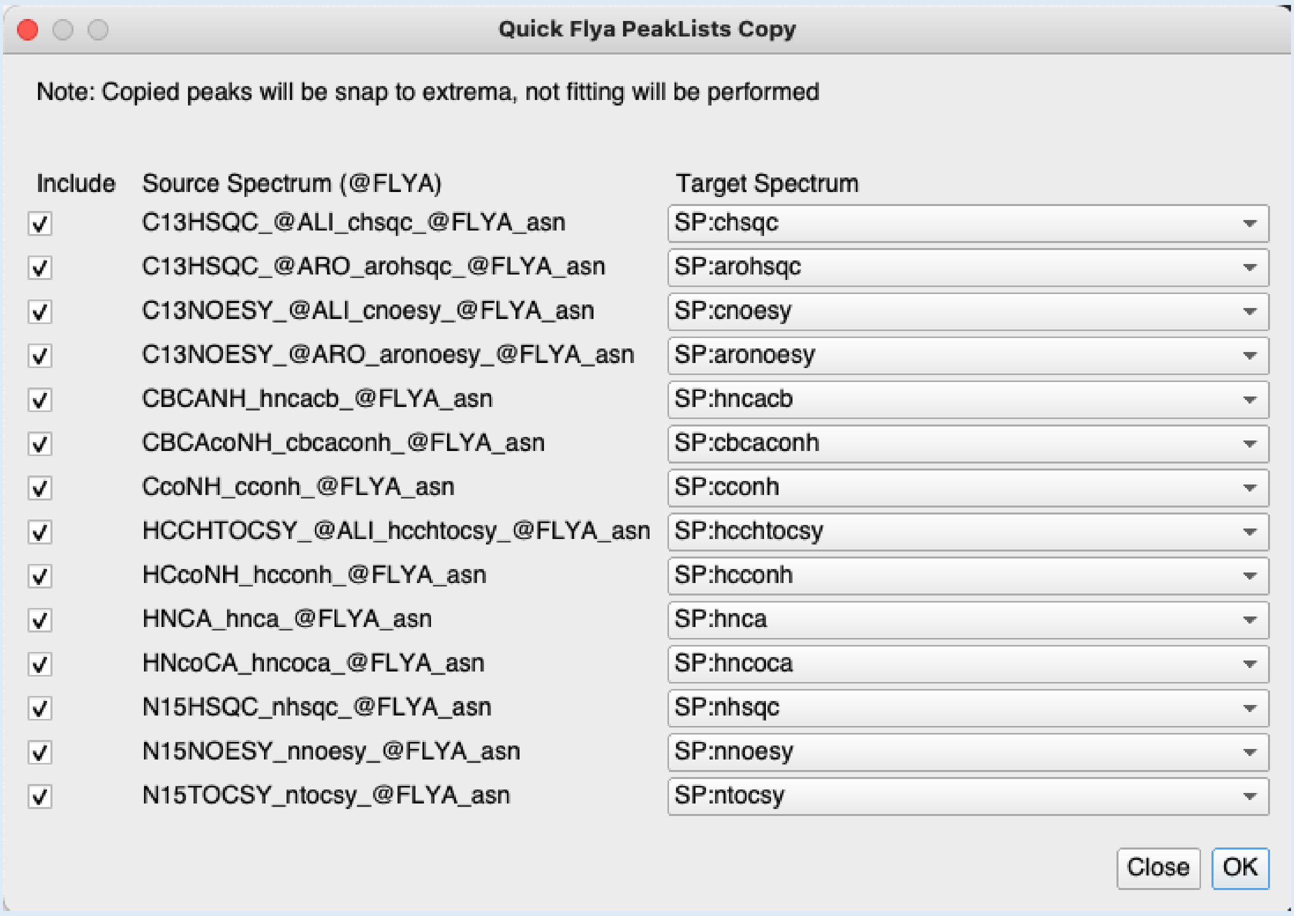


2A Copy peak list to the target spectra

- Drop the **SP:nhsqc** spectrum into the DropArea to display it.
- Expand the **SP:N15HSQC_nhsqc_@FLYA_asn** spectrum in the sidebar to view its peak lists.
- Drag and drop the **PL:N15HSQC_nhsqc_@FLYA_asn.1** peak list onto the SpectrumDisplay. This will open the **Copy PeakList** popup.
- Make sure the **SP:nhscq** spectrum is selected as the Target Spectrum.
- You can keep all the settings as they are or untick the **Recalculate volume** (as volumes are not needed)
- *Alternatively:* Go to **Spectrum > Copy PeakLists...**

2 Coping peak list between spectra

ubiquitin_spectra_2.ccpn



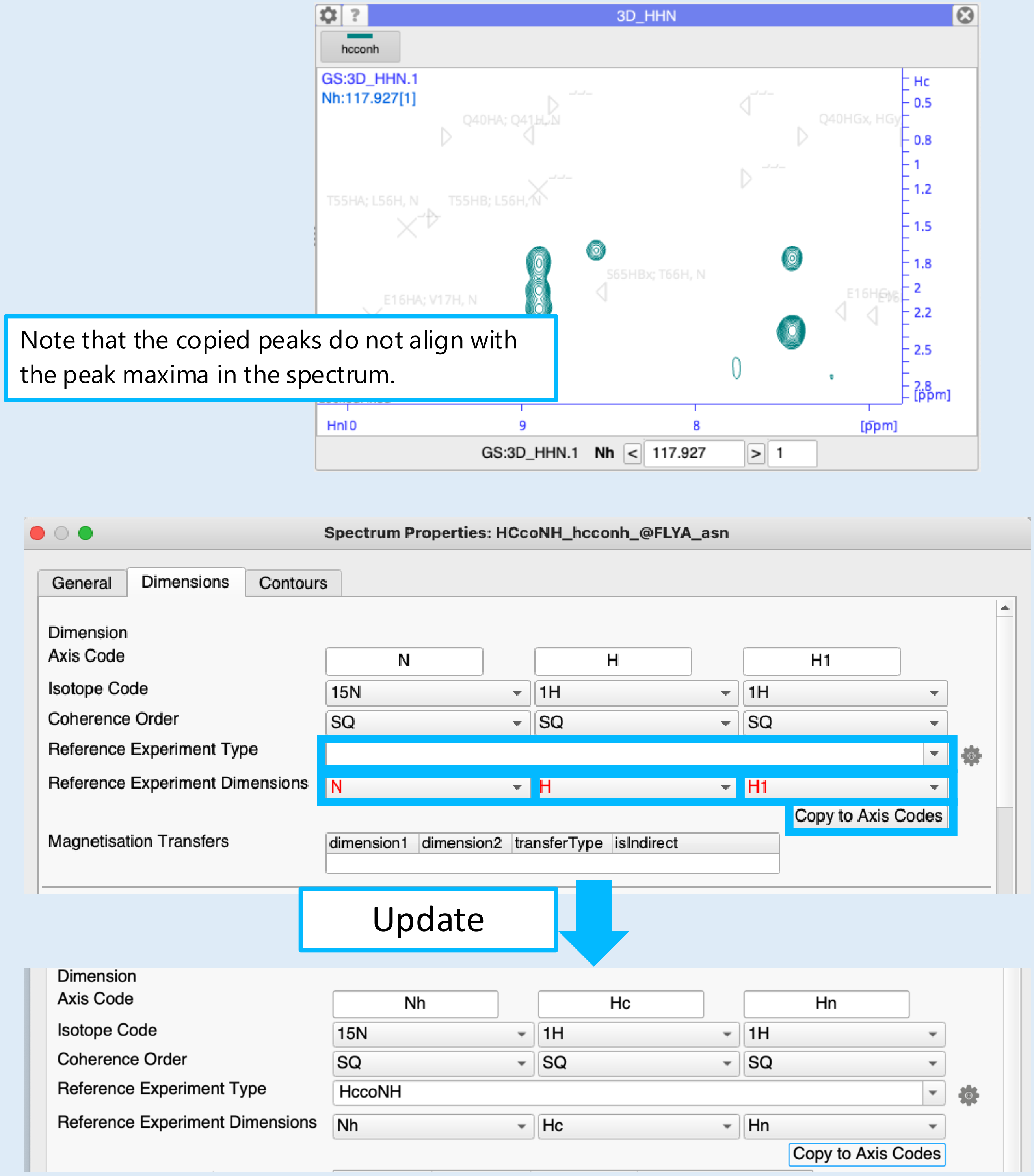
This macro automatically pairs spectra containing '@FLYA' in their names with matching target spectra in the project.

2B Copying multiple '@FLYA' peak lists to target spectra simultaneously

- Go to **Macro > Run CCPN Macros > quickFlyaCopy**
- Click **OK**
- Depending on the number of spectra it may take few minutes to run this macro.
- **Note:** only peak positions and assignments will be copied. If fitting is necessary, it needs to be performed separately on each peak list.
- *Known Issue:* you may get a message with a warning ("position constrain… violated by value…"), it ***should be*** harmless, we are working to resolve this.

3 Tips and suggestions

ubiquitin_spectra_2.ccpn



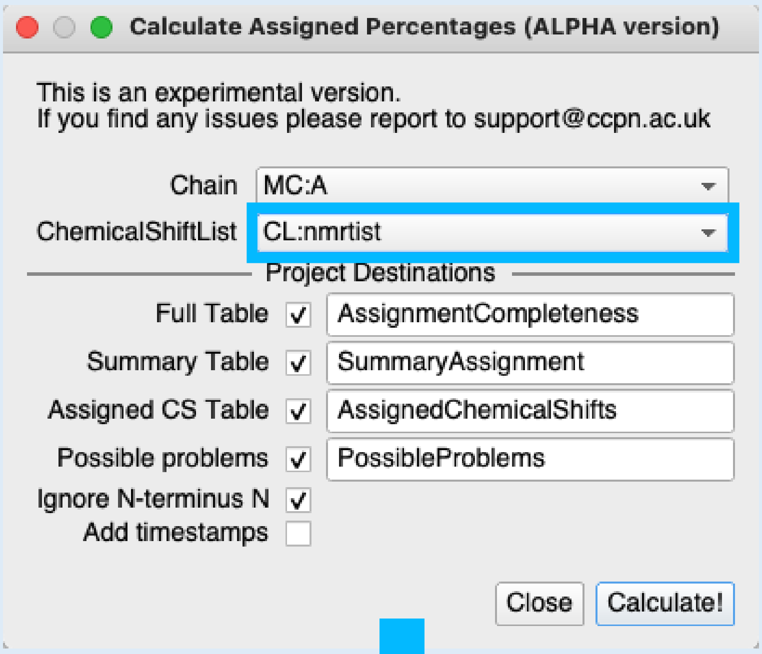
3A Inspecting copied peaks

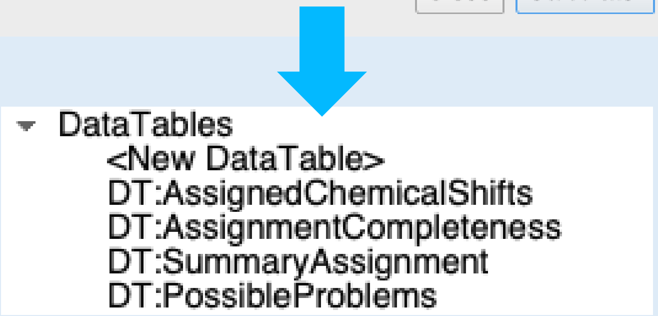
- Drag **SP:hcconh** onto drop area to display it. You can see that the peaks were not copied into the correct dimension.
- Open the **SP:HCcoNH_hcconh_@FLYA_asn Spectrum Properties** popup (double-click on the spectrum in the sidebar)
- Update **Reference Experiment Type** and **Reference Experiment Dimensions** and click on **Copy to Axis Codes**.

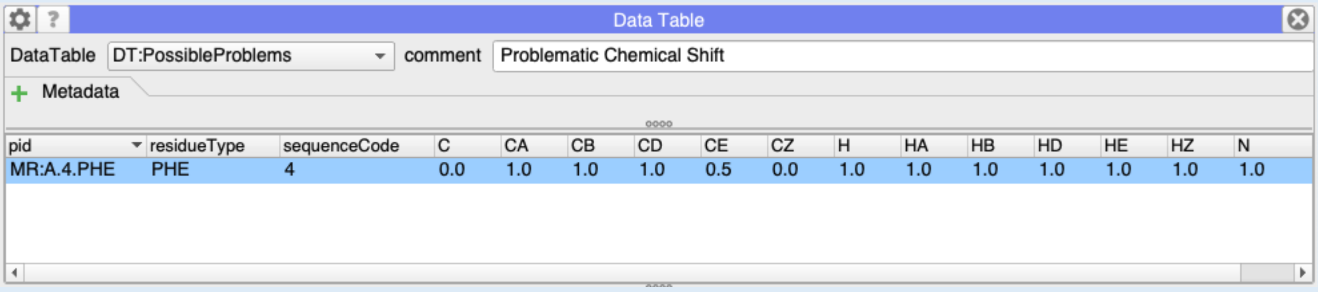
  Note: the order of the spectrum dimensions is different than it was in the original spectra!
- Copy the PeakList from **SP:HCcoNH_hcconh_@FLYA_asn** to **SP:hcconh** again

3 Tips and suggestions

ubiquitin_spectra_2.ccpn







This is very good quality data with a large number of spectra, consequently there will only be a very small number of errors.

3B Inspecting chemical shift list

- Go to **Macro > Run CCPN Macros > ProteinAssignmentStatistics**
- Select the **nmrtist** chemical shift list to be inspected, click **Calculate!**

  Four new DataTables will be created.
- Drag **DT:PossibleProblems** into the Drop Area to display it.

  You can see that **PHE.4** was flagged with an issue as only half the **CE** carbons are reported. The **PHE.4** chemical shifts and assignments should be inspected in detail.
- ***Note:*** It is common for Artina to assign aromatic ring atoms with separate shifts (non-degenerate, e.g. **HD1** and **HD2**), in most cases, where the shift difference is very small, those should be annotated as degenerate (**HD%**).

Contact Us

Website:
www.ccpn.ac.uk

Suggestions and comments:
support@ccpn.ac.uk

Issues and bug reports:
https://www.ccpn.ac.uk/forums/

Cite Us

Skinner, S. P. *et al.* CcpNmr AnalysisAssign: a flexible platform for integrated NMR analysis. *J. Biomol. NMR* (2016). doi:10.1007/s10858-016-0060-y

Cite NEF

Gutmanas, A. et al. NMR Exchange Format: A unified and open standard for representation of NMR restraint data. Nature Structural and Molecular Biology 22, 433–434 (2015).